Intro to C

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Intro to C

C is sadly a programming language. 😔

How to write hello world in C?

```
#include <stdio.h> // Standard Input/Output header

int main() {
printf("Hello, World!\n"); // Output text
return 0; // End program
}
```

Breakdown:

- `#include <stdio.h>` → imports functions like `printf`.
- `int main()` → entry point. Every C program **starts** from `main()`.
- `{ ... }` → body of the function (where instructions live).
- `printf(...)` → prints text to screen.
- `return 0;` → tells OS program ran successfully.

Variables and Data types

Type	Size (approx)	Example values
`int`	4 bytes	1, -100, 2025
`float`	4 bytes	3.14, -0.5
`double`	8 bytes	3.141592653589
`char`	1 byte	'a', 'Z', '@'
`_Bool`	1 byte	0 (false), 1 (true)

Operators

- **Arithmetic**: `+ - * / %`
- **Relational**: `== != > < >= <=`
- **Logical**: `&& || !`
- **Increment/Decrement**: `++ --`

Input and Outputs

- Input: `scanf`
- Output: `printf`

  `int num;`

  `printf("Enter a number: ");`

  `scanf("%d", &num); // & = address-of`

  `printf("You entered %d\n", num);`

Conditional Statements

```
if (condition) {
// code
} else if (another condition) {
// code
} else {
// code
}
```

Example

```
int marks = 85;
if (marks >= 90) printf("Grade: A+\n");
else if (marks >= 75) printf("Grade: A\n");
else printf("Keep trying!\n");
```

Week 1 Challenges

1. Write a program that takes two numbers and prints their sum, difference, product, and quotient.
2. Write a program that checks if a number is **even or odd**.

3. Write a program that asks for your **name** and prints:

   `Hello, <yourname>! Welcome to C programming.`
4. Write a program that calculates the **area of a circle** (input radius).
5. Write a program that prints a **multiplication table** for a given number.

Week 1 Answers

1. `#include <stdio.h>`

```
int main(){

int num1, num2;

printf("Enter the first number \n");

scanf("%d", &num1);

printf("Enter the second number \n");

scanf("%d", &num2);

printf("Sum = %d\n", num1 + num2);

printf("Difference = %d\n", num1 - num2);

printf("Product = %d\n", num1 * num2);

printf("Qoutient = %d\n", num1 / num2);

return 0;

}
```

2. `#include <stdio.h>`

```
int main(){

int num;

printf("Enter your number \n");

scanf("%d", &num);
```

```
if (num%2 == 0)
{
printf("Number is even \n");
}
else
{
printf("Number is odd\n");
}
return 0;
}
```

3. Write a program that asks for your **name** and prints:

   `Hello, <yourname>! Welcome to C programming.`
4. Write a program that calculates the **area of a circle** (input radius).
5. Write a program that prints a **multiplication table** for a given number.

References